

Run_55 Resist-HV Scan: MM Tracking vs Time-Since-Flash and the Search for an Operating Point

n_TOF EAR2 X17 — DREAM Micromegas, scint-doubles trigger, ^3He target (July 18–19, 2026)

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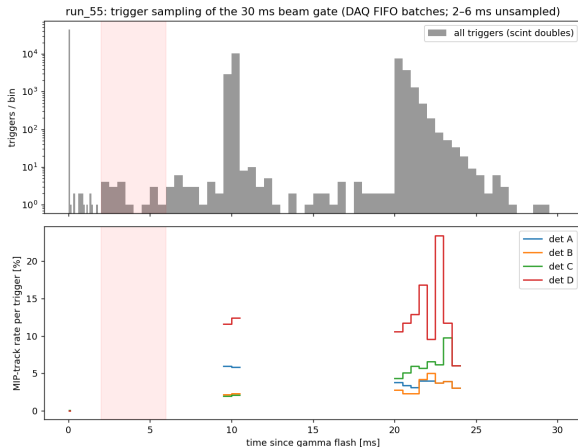
analysis and slides prepared with Claude (Anthropic AI assistant)

July 19, 2026

The scan: what ran, what survived

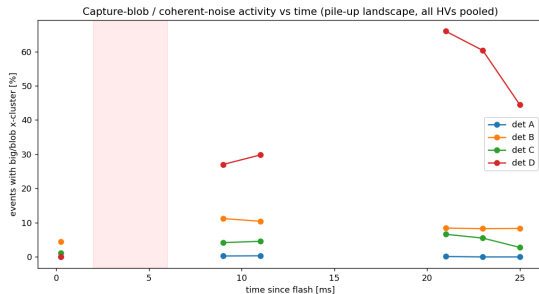
- ▶ **Cyclical resist scan** r560 $\rightarrow$ r520 V in -5 V steps (9 points), all four dets stepped together; 10 min/subrun.
- ▶ Drift fixed: **800 V** B/C/D, **600 V** A (A sparks on drift at 800; stable at 600, maybe 700 max).
- ▶ Gas Ar/iC₄H₁₀ 90/10; ³He target; no beam filter.
- ▶ Trigger: **scint DOUBLES** (≥ 2 -of-4 wall $\times$ plastic), 30 ms beam gate — MM-independent $\Rightarrow$ tracks-per-trigger is a valid efficiency proxy.
- ▶ DREAM: 32 smp $\times$ 60 ns window, latency 35, **no ZS**.
- ▶ DAQ machine died mid-run; data auto-synced to EOS. **24 subruns recovered** (3 cycles r535–560, 2 cycles r520–530); only the in-flight subrun lost.
- ▶ 76 214 triggers, ~ 190 beam pulses/subrun.
- ▶ Analysis: 25_hv_scan_extract.py / 25b_hv_scan_analysis.py; writeup HV_SCAN_RUN55_ANALYSIS.md.

Time axis: burst anchoring and the DAQ sampling comb



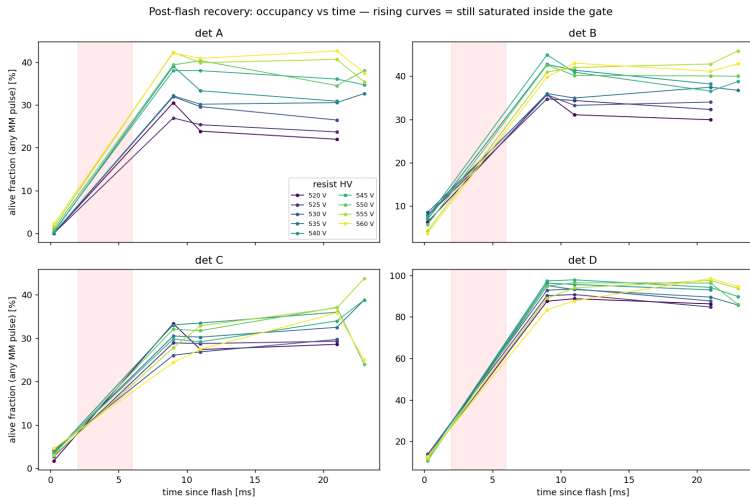
- ▶ One trigger burst per beam pulse (~ 3 s apart); **first trigger of a burst = gate open = γ flash (t_0)**. Valid for all bursts (batch structure identical whether the MM recorded the flash or railed empty).
- ▶ **No-ZS events are ~ 1.6 MB $\Rightarrow$ FIFO comb:** 4–5 events at 0–0.5 ms, dead ~ 8 ms, batch at **8–12 ms** (b1), batch at **16–28 ms** (b2).
- ▶ **2–6 ms after flash — including the 3 ms physics target — was never sampled.**

The ^3He capture flood (dominant systematic)



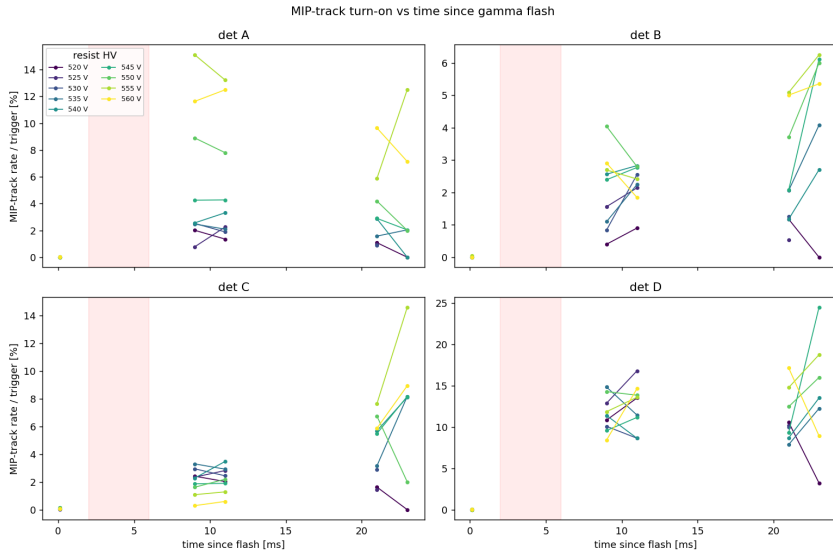
- ▶ 19 m flight path / 2200 m/s $\Rightarrow$ thermal neutrons arrive at **8.6 ms** — batch b1 sits ON the thermal peak; b2 is sub-thermal.
- ▶ $n + ^3\text{He} \rightarrow p + t$ (764 keV, heavily ionizing, mm–cm range): wide high-charge **blobs** in the MM windows.
- ▶ Det D: capture-scale cluster in $\gtrsim 90\%$ of late windows. Arms are equidistant (pinwheel) — D's $\sim 2\times$ gain is the likely reason it sees so many more.
- ▶ Blobs fake/mask tracks $\Rightarrow$ keep top-5 clusters/plane; “MIP-like” = 3–20 strips, ≤ 25 mm, x/y sample-coherent.

Q1a — turn-on: dead at the flash, everywhere

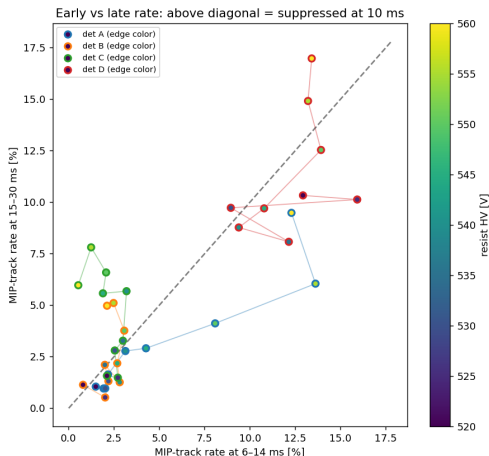


0–0.5 ms: 0 tracks in 44 686 triggers; occupancy 0–7% (D ~12% = its capture flood).
Scint trigger rate itself is flat across the gate — this is purely an MM outage. **Rising curves (C,**

Q1b — MIP-track rate vs time, per HV



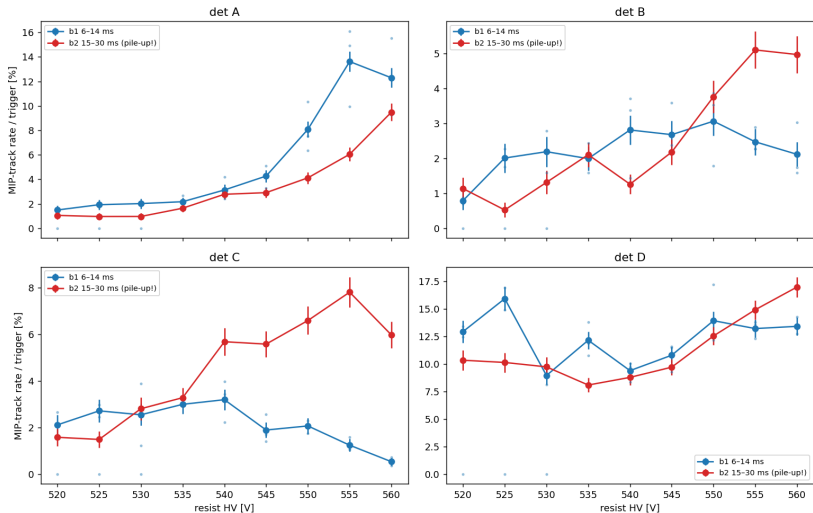
Q1c — saturation time GROWS with HV



- ▶ ≤ 550 V: rates and occupancy flat b1→b2 — recovered before 8 ms at every det.
- ▶ **C at 555–560 V**: 0.5–1.2% at 8–12 ms vs 5.9–7.8% at >15 ms (points far *above* the diagonal) — gain still sagged at 10 ms.
- ▶ **D at 560 V**: same signature, milder (occupancy 83%→98% through the gate).
- ▶ **A**: points *below* the diagonal — efficiency *declines* through the gate (13.6%→6.1% at 555 V). Open; candidate: space charge in the 600 V drift under thermal load.
- ▶ Early amplitude “suppression” appears as **rate loss + survivor bias** (only the Landau tail clears threshold), not low medians.

Q2a — efficiency proxy vs resist HV (the money plot)

Efficiency proxy vs resist HV (dots: single cycles, b1)



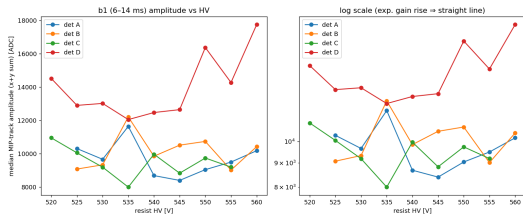
Q2b — reading the efficiency curves

MIP-track rate per trigger at 8–12 ms [%]
(geometric ceiling per arm $\sim 50\%$, doubles trigger):

HV	A	B	C	D*
520	1.5	0.8	2.1	12.9
530	2.0	2.2	2.5	8.9
540	3.1	2.8	3.2	9.4
550	8.1	3.1	2.1	13.9
555	13.6	2.5	1.2	13.2
560	12.3	2.1	0.5	13.4

- ▶ **A**: clean rise, **no plateau by 560 V**, and NO saturation sag even at 560 — simply under-biased.
- ▶ **B**: weak response 2–3%; late-gate rises to $\sim 5\%$ at 555–560. Complicated by its wide (~ 200 mm) low-amp ringing clusters. Inconclusive.
- ▶ **C**: true gain peak ≥ 555 V (late-gate), but long-saturation penalty above 550.
- ▶ **D***: contaminated by capture fragments at low HV (13% at 520 V is NOT MIP efficiency); plateau-ish 545–560.

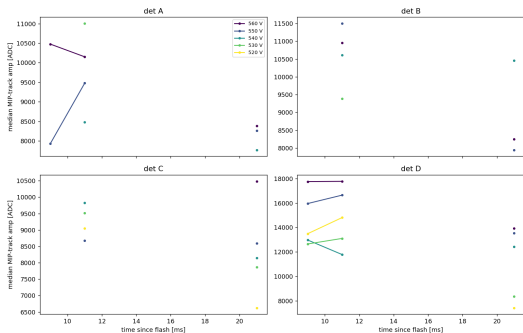
Amplitudes: medians are threshold-biased; spectra + gain notes



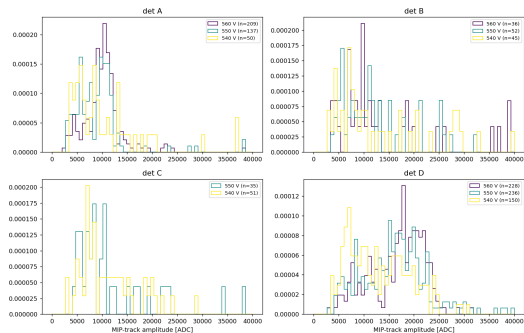
- ▶ Median MIP-track amp sits at 8–12k ADC (x+y) at every HV — only tracks *above detection* enter, so this is NOT a gain curve.
- ▶ Real gain slope from D's resist current: $1.0 \rightarrow 5.4 \mu\text{A}$ over $520 \rightarrow 560 \text{ V} \Rightarrow$ **gain doubles every $\sim 17 \text{ V}$** . (A's current is leakage-dominated: $2.7 \rightarrow 3.7 \mu\text{A}$.)
- ▶ Where sag exists, b1 medians are biased *high* (ratio up to 1.3–1.4 vs b2) — survivor bias, consistent with suppressed gain early.

Amplitude vs time and b1 spectra

MIP-track amplitude vs time since flash (b2 ≥ 15 ms carries capture pile-up, esp. det D)

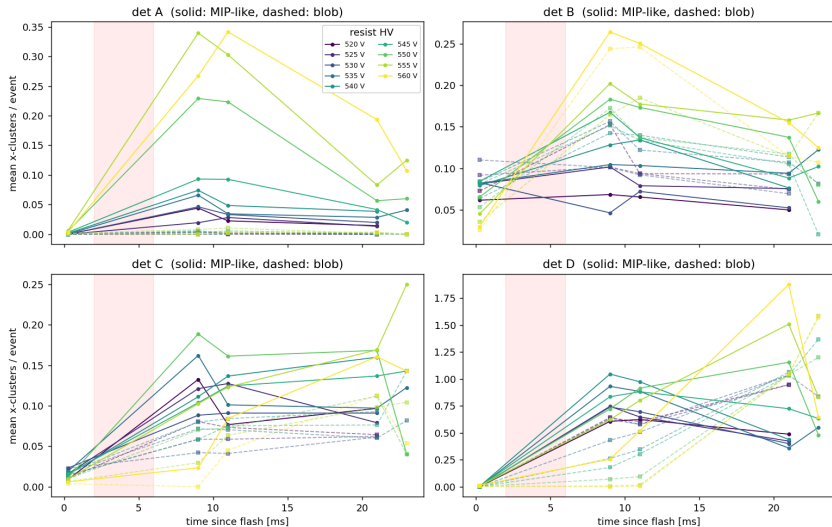


b1 MIP-track amplitude spectra

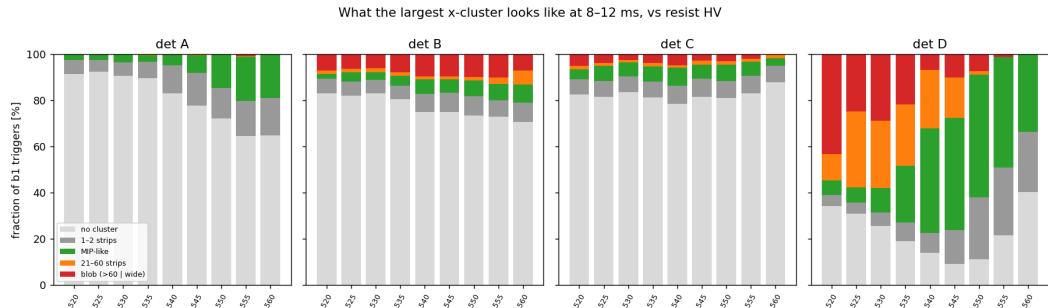


Cluster multiplicity vs time and HV

Cluster multiplicity vs time — MIP-like (3-20 strips, ≤ 25 mm) vs blob (wider/bigger) x-clusters



Cluster composition at 8–12 ms: four different detectors



A: MIP fraction grows cleanly with HV, almost no blobs. B/C: modest MIP fractions + persistent blob/ringing background; C's dead fraction *grows* at 555–560 (sag). D: capture blobs (red/orange) at low HV morph into MIP-class + “no cluster” at 555–560 — FEU saturation fragmentation + gain sag.

Recommendation (for ≥ 3 ms operation, this gas/config)

det	operating point	why
A	≥ 560 V, scan higher	rate still rising at 560; no sag even at 560. Drift capped at 600–700 V (sparks at 800) — consider pushing drift to 700 as well
B	550–555 V (tentative)	late-gate gain still rising to 555; no strong sag; blocked on the ringing/wide-cluster issue
C	545–550 V	gain wants ≥ 555 but saturation >10 ms there; 545–550 = best early-gate efficiency
D	540–550 V	plateau by ~ 545 ; sag at 560; full capture visibility everywhere

Caveat: 3 ms itself was never sampled. At 555+ V, C/D are provably NOT recovered at 8–12 ms — the 3 ms goal likely requires the lower operating points, or a saturation fix.

Open items / follow-up wanted

- ▶ **Sample 1–8 ms**: rerun with zero-suppression ON (or short window) to shrink the ~ 8 ms readout dead time, or step a delayed trigger gate through 1–8 ms. Only way to answer “alive at 3 ms”. (*Waiting on DAQ machine repair.*)
- ▶ **Det A in-gate decline** ($b1 \rightarrow b2$): dedicated look; space charge in the 600 V drift is the prime suspect — test at drift 700 V.
- ▶ **Det B ringing / wide clusters**: waveform-level diagnosis (needs `raw_daq_data` or `hits_root`, still on EOS).
- ▶ Higher-HV point for A (565–575 V) to find its plateau.
- ▶ Absolute per-arm efficiency: tag-and-probe across arms from the doubles trigger (16-pattern likelihood; machinery in `ntof_tracking/`).